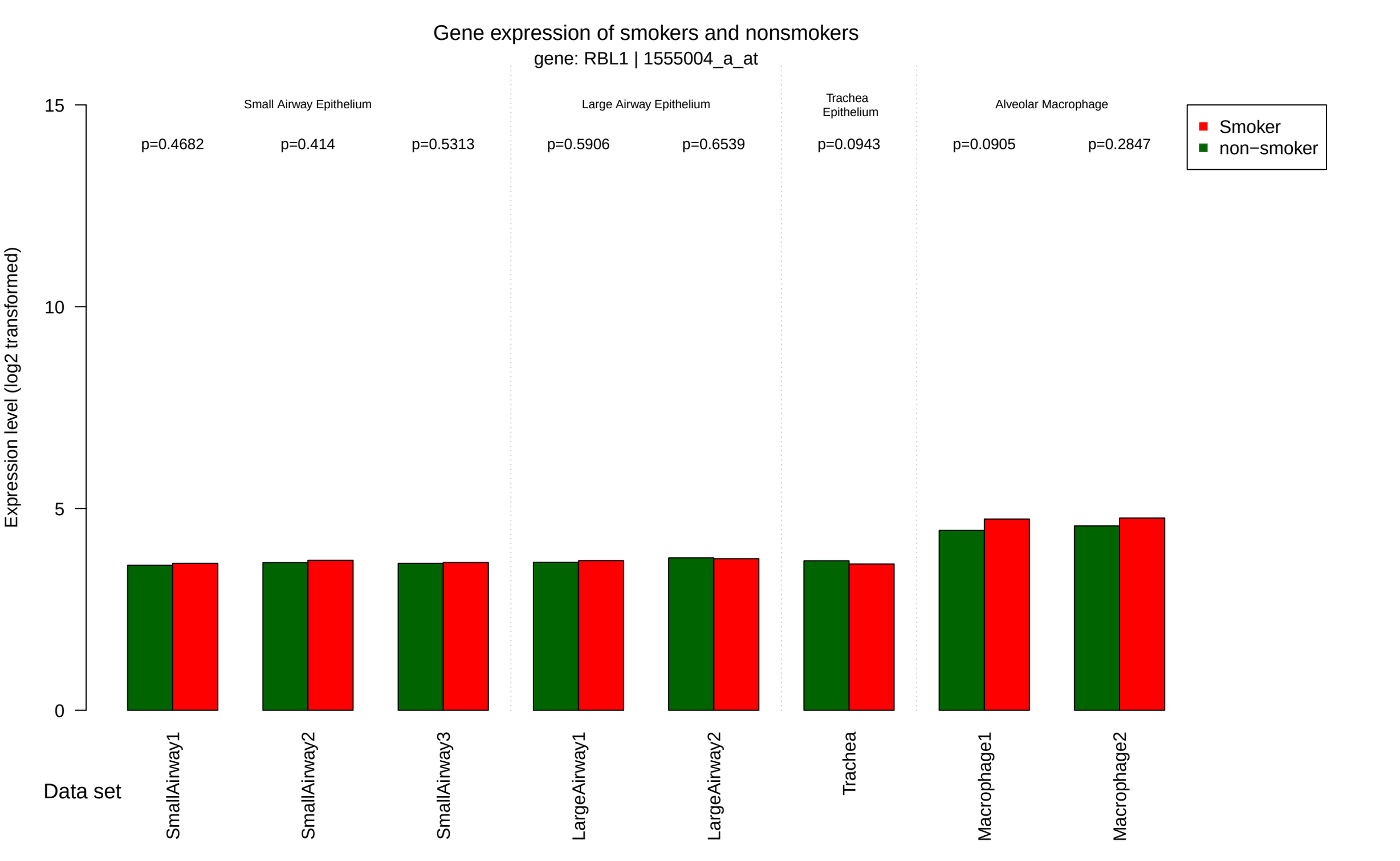
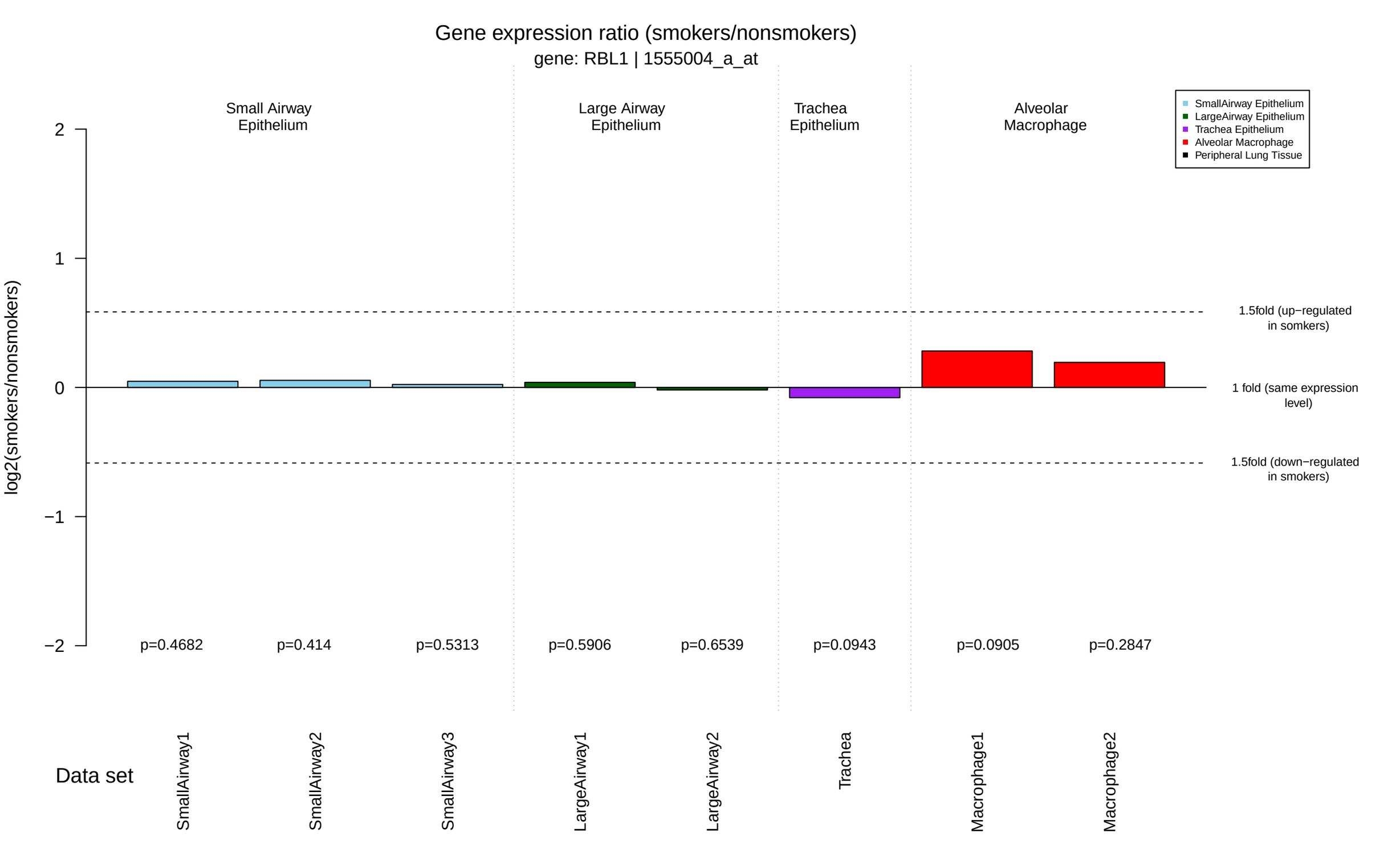
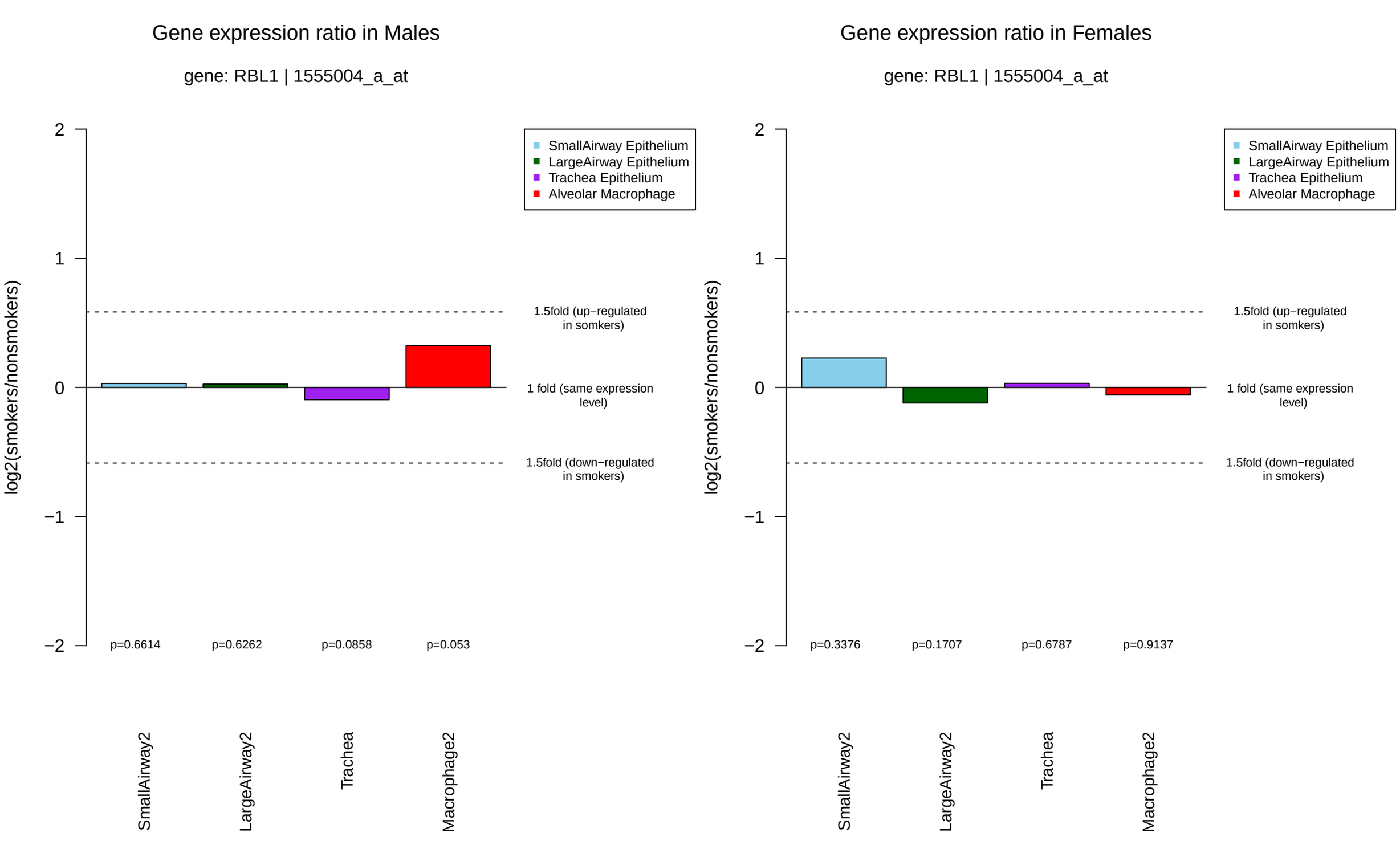
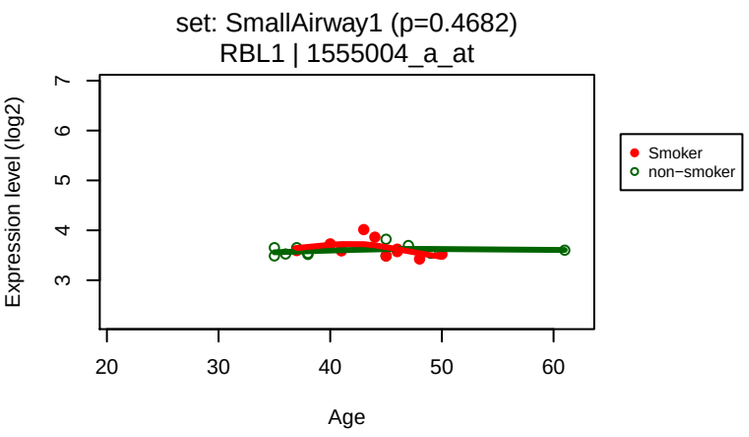
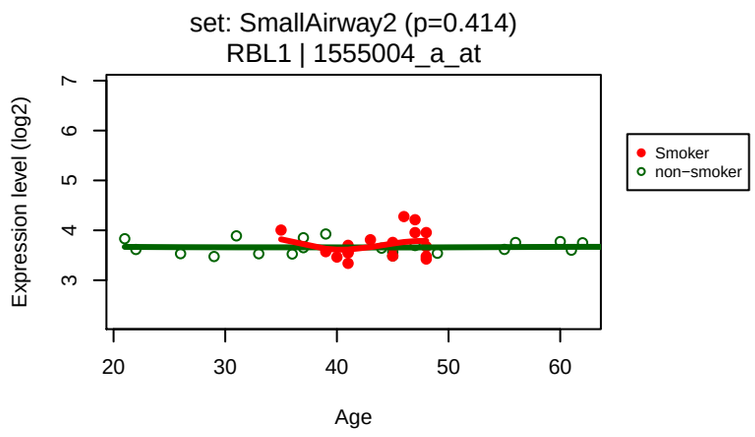
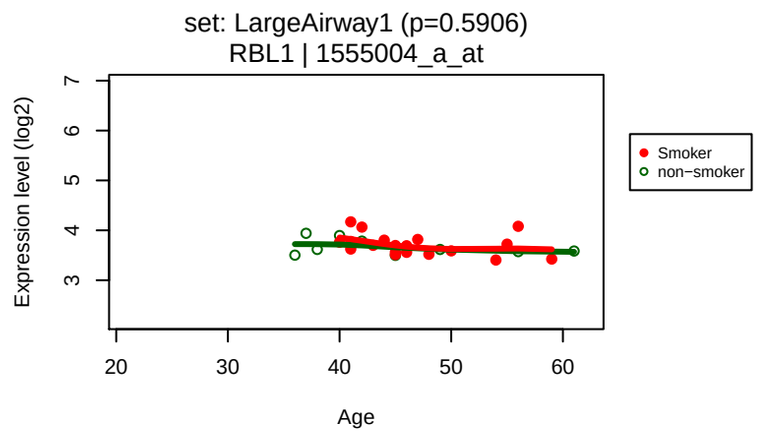

Gene expression of smokers and nonsmokers
gene: RBL1 | 1555004_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4682
p=0.414
p=0.5313
p=0.5906
p=0.6539
p=0.0943
p=0.0905
p=0.2847
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

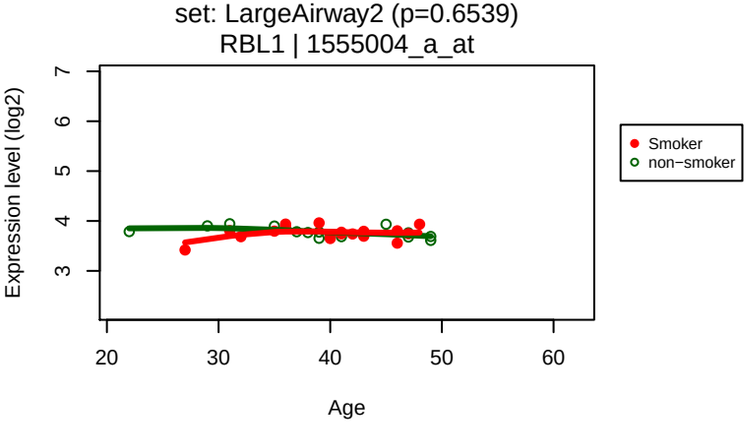

Gene expression ratio (smokers/nonsmokers)
gene: RBL1 | 1555004_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4682
p=0.414
p=0.5313
p=0.5906
p=0.6539
p=0.0943
p=0.0905
p=0.2847
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

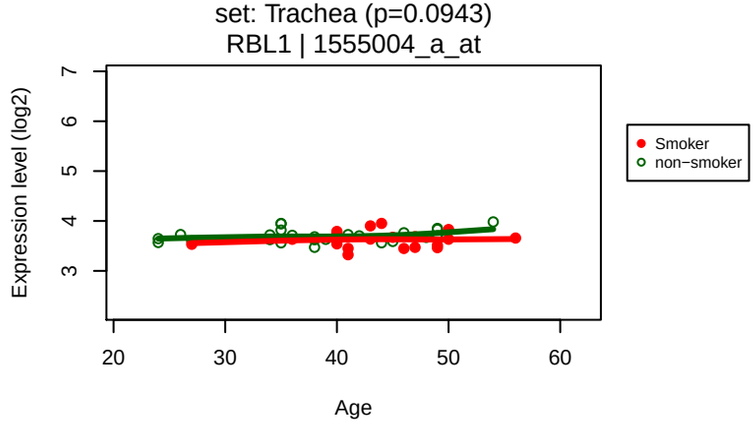

Gene expression ratio in Males
gene: RBL1 | 1555004_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6614
p=0.6262
p=0.0858
p=0.053
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RBL1 | 1555004_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3376
p=0.1707
p=0.6787
p=0.9137
SmallAirway2
LargeAirway2
Trachea
Macrophage2

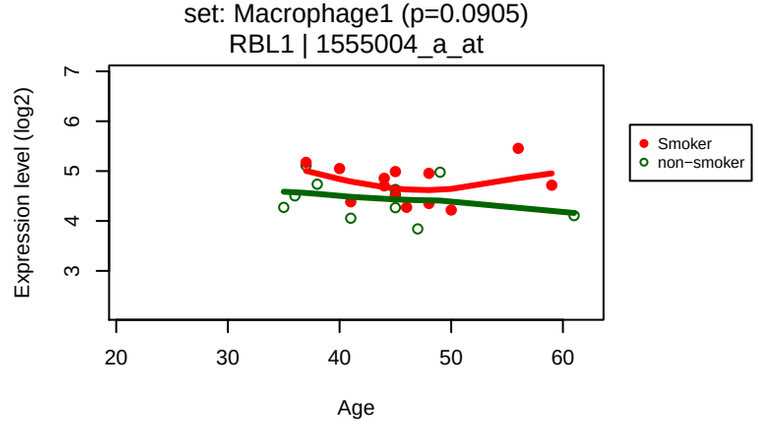

set: SmallAirway1 (p=0.4682)
RBL1 | 1555004_a_at
Expression level (log2)
Age
Smoker
non−smoker

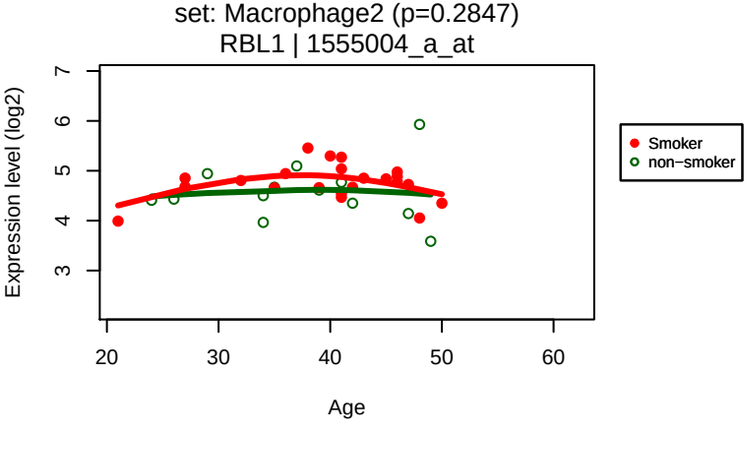

set: SmallAirway2 (p=0.414)
RBL1 | 1555004_a_at
Expression level (log2)
Age
Smoker
non−smoker

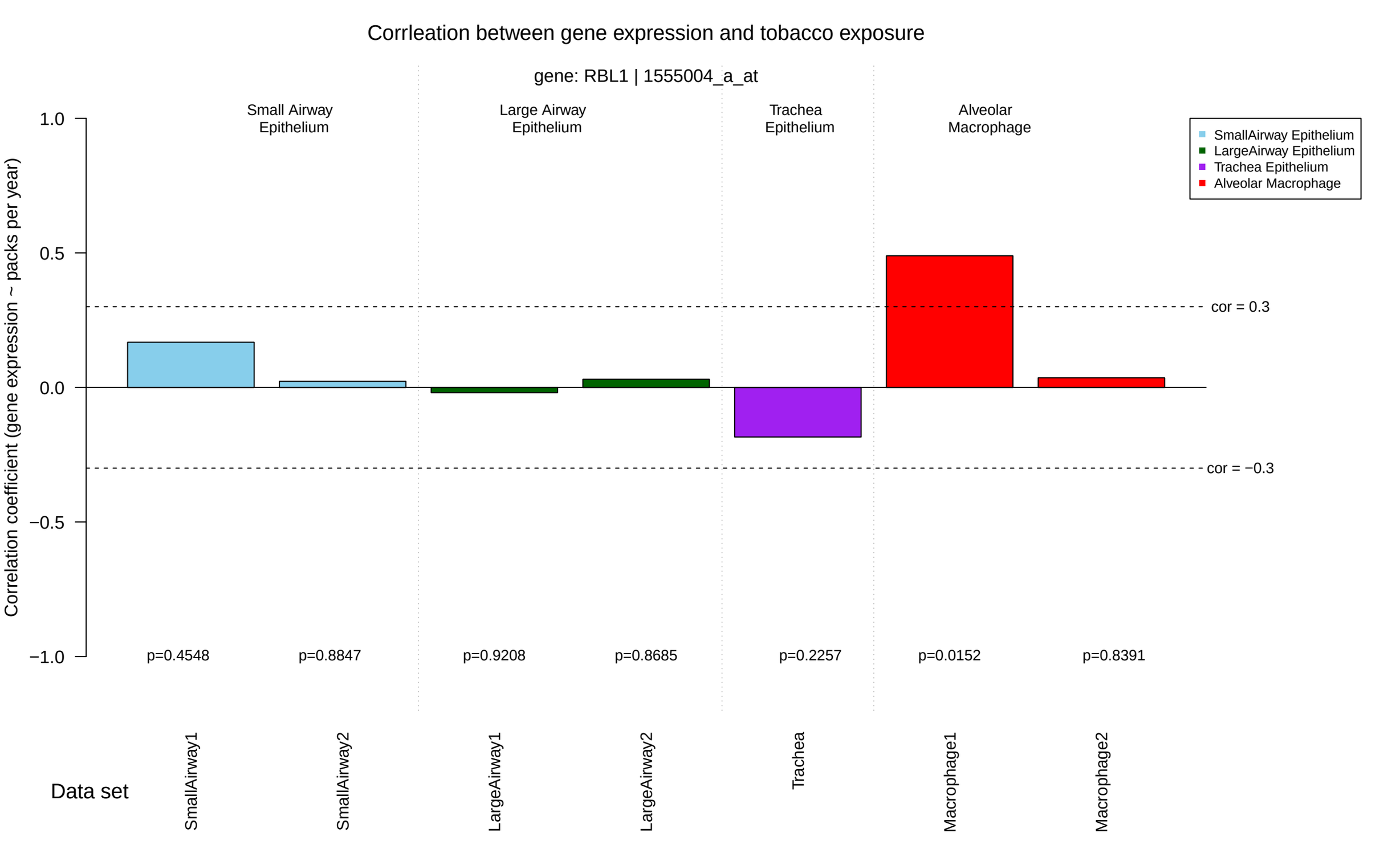

set: LargeAirway1 (p=0.5906)
RBL1 | 1555004_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6539)
RBL1 | 1555004_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0943)
RBL1 | 1555004_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0905)
RBL1 | 1555004_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2847)
RBL1 | 1555004_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RBL1 | 1555004_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4548
p=0.8847
p=0.9208
p=0.8685
p=0.2257
p=0.0152
p=0.8391
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2